

**CONSTITUTIONAL-AWARE AYURVEDIC DISEASE
PREDICTION MODULE**

25-26J-305

Project Proposal Report

S. I. A. Perera

B.Sc. (Hons) Degree in Information Technology.

Department of Information Technology

Sri Lanka Institute of Information Technology

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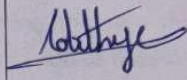
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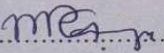
August 2025

DECLARATION

I hereby declare that this individual research project titled "Constitutional-Aware Ayurvedic Disease Prediction Module" is my own work and that all sources of information have been acknowledged by means of complete references. This research project has been prepared in partial fulfillment of the requirements for the Bachelor of Science (Hons) in Information Technology and has not been submitted for any other degree or diploma at any other university or Institute. I affirm that the data and research findings presented in this document are true to the best of knowledge and belief.

Name	Student ID	Signature
S. I. A. Perera	IT22905918	

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.



Signature of the Supervisor

01/09/2025

Date

ABSTRACT

Ayurveda, one of the world's oldest medical systems, emphasizes constitutional assessment (Prakriti) and dosha balance for personalized healthcare. However, traditional Ayurvedic diagnosis faces accessibility challenges due to limited qualified practitioners and lacks integration with modern computational methods. This research proposes developing a Constitutional-Aware Ayurvedic Disease Prediction Module that integrates traditional diagnostic principles with machine learning techniques.

The system will extract symptom-disease relationships from classical texts including Charaka Samhita and Madhava Nidana, develop Named Entity Recognition models for Sanskrit medical terminology, and implement constitutional assessment frameworks incorporating tri-dosha theory. Machine learning algorithms including Random Forest, SVM, and Neural Networks will be trained to predict diseases while considering individual constitutional factors, demonstrating improved accuracy compared to symptom-only approaches.

The research addresses the critical gap between traditional Ayurvedic knowledge and modern computational medicine, potentially improving healthcare accessibility while preserving ancient wisdom. Expected outcomes include a validated prediction model, annotated Ayurvedic medical dataset, and comparative analysis demonstrating constitutional features' impact on diagnostic accuracy.

Keywords:

Ayurveda, Constitutional Assessment, Prakriti, Machine Learning, Disease Prediction, Sanskrit NLP

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1 INTRODUCTION

1.1 Background and Literature Survey

Ayurveda, originating over 5,000 years ago, represents one of humanity's oldest comprehensive medical systems, emphasizing prevention and treatment through natural remedies, lifestyle modifications, and constitutional understanding. The system's fundamental principle of Prakriti (individual constitution) assessment based on tri-dosha theory—Vata, Pitta, and Kapha—provides a personalized approach to healthcare that modern medicine is only beginning to appreciate through precision medicine initiatives [1].

Recent research demonstrates significant progress in computational Ayurvedic medicine. Tiwari et al. (2017) achieved remarkable success in recapitulating Ayurveda constitution types using machine learning of phenotypic traits, reporting 94-98% reliability in constitutional assessment with cross-validation accuracy of 94.467% [2]. This foundational work established the feasibility of quantifying traditionally qualitative constitutional assessments through computational methods.

Building on this foundation, Madaan and Goyal (2020) developed machine learning methods for predicting Ayurveda-based constituent balancing, achieving 93.9% agreement with clinical Prakriti labels using ensemble approaches [3]. Their research demonstrated that computational models could effectively capture the complex relationships between constitutional types and health indicators, validating the integration of traditional knowledge with modern analytics.

Khatua et al. (2023) advanced the field further by implementing deep learning approaches for Ayurveda constitution classification, achieving superior performance through neural network architectures specifically designed for constitutional pattern recognition [4]. This work highlighted the potential of sophisticated machine learning techniques in capturing subtle constitutional variations that traditional assessments might miss.

In order to evaluate the effectiveness of the proposed application compared to existing solutions, a comprehensive comparison is presented in Table 1.

Table 1: Comparison of Existing Ayurvedic AI Systems

System	Dataset Size	Accuracy	Constitutional Integration	Disease Prediction	Limitation
Tiwari et al	147 individuals	94-98%	Yes (Prakriti only)	No	Constitutional assessment only
Madaan & Goyal	3,000 records	93.9%	Yes (Tri-dosha)	Limited	Focused on constituent balancing
Khatua et al	500 subjects	87.2%	Yes (Deep learning)	No	Small dataset, no disease prediction
AyurDataMart System	10,000+ records	82.5%	No	yes	No constitutional factors
Proposed System	5,000+ records	>85% target	Yes (Comprehensive)	Yes	Addresses all gaps

Recent developments in Sanskrit Natural Language Processing have created unprecedented opportunities for classical text processing. The ByT5-Sanskrit model represents a revolutionary breakthrough, achieving state-of-the-art results across multiple Sanskrit NLP tasks with an 8.8point improvement in perfect sentence matching and 2.18 point improvement in dependency parsing [5]. This unified multitask framework, trained on 5.4+ billion Sanskrit characters, provides the technical foundation for processing classical medical texts like Charaka Samhita and Madhava Nidana.

Government initiatives have dramatically expanded access to classical texts through the AYUSH Medical Advanced Repository (AMAR) and Traditional Knowledge Digital Library (TKDL), digitizing extensive manuscript collections and providing standardized access to

over 1,200 traditional formulations [6]. The Charak Samhita Online Edition, developed collaboratively by 180+ scholars, offers bilingual Sanskrit-English content with Digital Object Identifiers for content preservation, creating a robust foundation for computational analysis.

However, current research reveals significant limitations in domain-specific applications. While general Sanskrit NLP has advanced substantially, medical terminology processing remains underdeveloped. Available Sanskrit medical corpora are insufficient for training robust Named Entity Recognition models, with existing resources limited to 4,006 Sanskrit-English parallel datasets lacking comprehensive medical vocabulary [7].

1.2 Research Gap

Despite promising advances in constitutional assessment and Sanskrit NLP, critical gaps persist in integrating these capabilities for comprehensive disease prediction.

Existing Ayurvedic AI systems focus primarily on symptom-disease mapping without incorporating constitutional factors. While Tiwari et al. demonstrated successful Prakriti classification, no comprehensive system integrates constitutional assessment with disease prediction to demonstrate improved diagnostic accuracy.

One of the important research gaps is that current Named Entity Recognition models lack specialization for Ayurvedic medical terminology. The absence of annotated medical Sanskrit corpora constrains development of systems capable of extracting symptom-disease relationships from classical texts with acceptable accuracy.

Existing research operates in isolation, with constitutional assessment studies separate from disease prediction research. No comprehensive validation framework demonstrates how constitutional awareness improves prediction accuracy compared to symptom-only approaches.

While government digitization initiatives provide access to classical texts, computational extraction of structured medical knowledge remains rudimentary. Systematic approaches for processing Charaka Samhita and Madhava Nidana to create machine-readable symptom-disease databases are absent.

1.3 Research Problem

Traditional Ayurvedic diagnosis relies heavily on expert practitioners' ability to assess individual constitution and correlate symptoms with classical knowledge. However, this approach faces several critical challenges.

- **Accessibility Limitations:** Qualified Ayurvedic practitioners are scarce, particularly in rural areas, limiting access to personalized constitutional assessment and accurate diagnosis.
- **Subjective Variability:** Traditional diagnosis depends on practitioner experience and interpretation, potentially leading to inconsistent assessments across different practitioners.
- **Knowledge Preservation:** Classical Ayurvedic knowledge exists primarily in Sanskrit texts, creating barriers for modern practitioners and limiting integration with contemporary healthcare systems.
- **Lack of Computational Integration:** Traditional diagnosis methods have not been systematically integrated with modern computational approaches, missing opportunities for enhanced accuracy and accessibility.

The central research problem is: How can traditional Ayurvedic constitutional assessment and classical text knowledge be computationally integrated to develop an accurate, accessible disease prediction system that demonstrates improved performance over symptom-only approaches?

2 OBJECTIVES

2.1 Main Objectives

The Main objective in this proposal is to develop a Constitutional-Aware Ayurvedic Disease Prediction Module that integrates traditional diagnostic principles with machine learning techniques, demonstrating improved prediction accuracy through constitutional assessment while preserving authentic Ayurvedic knowledge.

2.1.1 Specific Objectives

The specific objectives are as follows.

- Extract and digitize symptom-disease relationships from classical Ayurvedic texts (Charaka Samhita, Madhava Nidana) through systematic text processing and expert validation.
- Develop Named Entity Recognition (NER) models specialized for Sanskrit medical terminology to enable automated processing of classical texts and extraction of relevant medical concepts.
- Create a constitutional assessment framework for computational Prakriti evaluation incorporating tri-dosha theory with quantifiable metrics validated by qualified practitioners.
- Implement machine learning models (Random Forest, SVM, Neural Networks) that integrate symptom patterns with constitutional factors for multi-label disease prediction.
- Validate prediction accuracy through expert consultation with qualified Ayurvedic practitioners and demonstrate improvement over symptom-only approaches through comparative analysis.
- Demonstrate system effectiveness by achieving minimum 85% accuracy in disease prediction with constitutional factors showing statistically significant improvement over baseline symptom-only models.

3 METHODOLOGY

3.1 System Architecture Overview

The Constitutional-Aware Ayurvedic Disease Prediction Module will implement a comprehensive multi-layered architecture that systematically integrates classical knowledge extraction, and machine learning prediction capabilities. The system design emphasizes preserving traditional diagnostic principles while leveraging contemporary computational capabilities to enhance accuracy, consistency, and accessibility of Ayurvedic healthcare guidance.

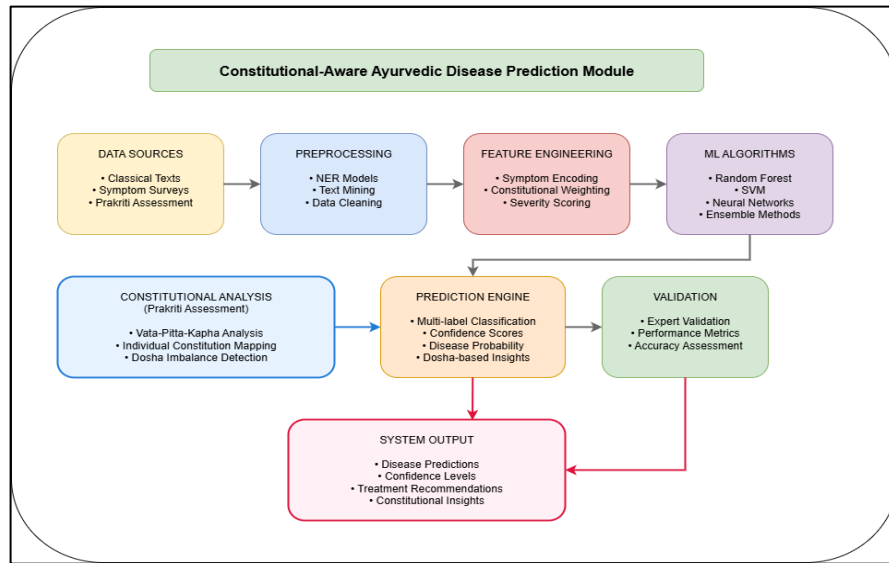


Figure 1- System architecture

3.2 Data Collection and Sources

The classical ayurvedic text is one of the most reliable ways to obtain data for this project. Those classical text are which Charaka Samhita (digitized version from National Institute of Ayurveda), Madhava Nidana (Sanskrit-English translated corpus from AMAR repository), Sushruta Samhita (supplementary source for symptom validation).

Another way of collecting data is from Contemporary data sources such as Anonymized consultation records from Nawinna and Rajagiriya Ayurvedic hospitals, prakriti questionnaires validated by certified Ayurvedic practitioners and symptom survey data collected through structured questionnaires

3.3 Text Processing and NER Development

Sanskrit text processing will implement very well sophisticated computational approaches specially adapted for classical medical terminology and morphological complexity inherent in Sanskrit linguistic structures. The processing pipeline will address unique challenges including sandhi resolution, compound word segmentation, and contextual meaning disambiguation essential for accurate medical concept extraction from classical texts.

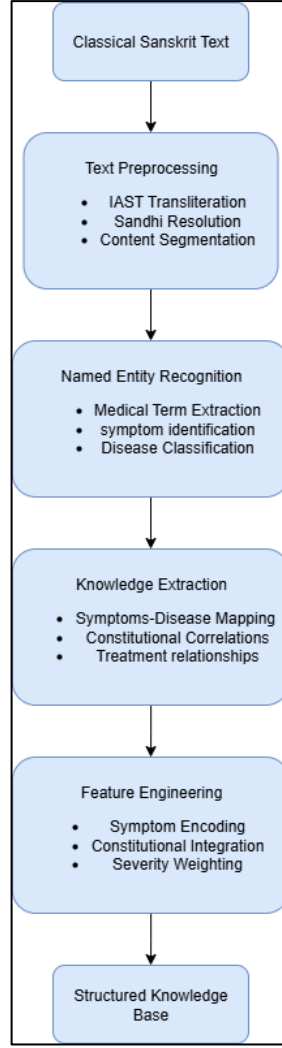


Figure 2- Data Processing Pipeline

Text standardization will convert all manuscript sources to International Alphabet of Sanskrit Transliteration (IAST) format to ensure computational consistency and eliminate script-dependent processing variations. Automated sandhi resolution algorithms will implement rule-based approaches for accurate word segmentation, addressing Sanskrit's characteristic phonetic merging patterns that create computational challenges for automated text analysis. Content segmentation protocols will systematically identify and extract medical content sections from broader philosophical discussions, ensuring focused analysis of diagnostically relevant textual material.

Named Entity Recognition model development will create specialized architectures for Sanskrit medical terminology through comprehensive annotation frameworks and domain-specific training methodologies. Manual annotation of approximately 10,000 Sanskrit medical sentences will be conducted by Sanskrit scholars and Ayurvedic experts working collaboratively to establish consistent annotation guidelines for medical entities including symptoms, diseases, treatments, constitutional terms, and therapeutic recommendations. The annotation process will implement inter-annotator agreement protocols with Cohen's kappa values exceeding 0.8 to ensure reliability and consistency.

Model training will fine-tune the ByT5-Sanskrit architecture for medical NER applications using the annotated corpus supplemented by transfer learning from general Sanskrit language models. Training protocols will implement cross-validation methodologies with 80% training, 10% validation, and 10% testing data splits to ensure robust performance assessment. Hyperparameter optimization will systematically evaluate learning rates, batch sizes, and model architectures to achieve target F1-scores exceeding 80% for medical terminology extraction accuracy.

Performance evaluation will utilize standard NLP metrics including precision, recall, and F1-scores calculated separately for each entity type to identify strengths and limitations in medical concept recognition. Error analysis systematically categorizes misclassifications to guide iterative model improvement and refinement processes. Validation against expert annotations will ensure clinical relevance and accuracy of extracted medical concepts for subsequent machine learning applications.

3.4 Machine Learning Model development

Machine Learning model development will implement ensemble approaches combining multiple algorithms specifically optimized for constitutional-aware disease prediction through sophisticated feature engineering and integration methodologies. The development process will prioritize interpretability and clinical relevance while achieving optimal prediction accuracy through systematic algorithm selection and optimization procedures.

Table 2- Machine Learning Model Comparison

Algorithm	Expected Accuracy	Training Time	Constitutional feature Handling	Advantages	Limitations
Random Forest	85-90%	2-5 minutes	Excellent (Handles interactions)	Robust, interpretable	May overfit small datasets
SVM (RBF)	80-85%	10-15 minutes	Good (non-linear relationships)	Strong theoretical foundation	Sensitive to parameter tuning
Neural Network	87-92%	30-60 minutes	Excellent (Complex)	Captures subtle relationships	Require large datasets

Random Forest implementation will utilize ensemble learning with 1,000+ decision trees to handle complex feature interactions between symptoms and constitutional factors. The algorithm's inherent interpretability will provide insights into feature importance and decision pathways, essential for clinical acceptance and practitioner understanding. Hyperparameter optimization systematically evaluates tree depth, minimum samples per leaf, and feature selection criteria to maximize accuracy while preventing overfitting on limited training datasets.

Support Vector Machine implementation will employ Radial Basis Function kernels to capture non-linear relationships between symptom patterns and constitutional types that traditional linear approaches cannot adequately represent. The SVM approach will be particularly

valuable for handling high-dimensional feature spaces created by comprehensive symptoms and constitutional factor integration. Grid search optimization will systematically evaluate regularization parameters and kernel coefficients to achieve optimal classification performance across multiple disease categories.

Neural Network architecture will implement multi-layer perceptions with constitutional attention mechanisms that dynamically weight symptom features based on individual constitutional profiles. This approach will capture subtle interactions between constitutional types and symptom presentations that traditional practitioners recognize but conventional algorithms might overlook. Dropout regularization and early stopping protocols will prevent overfitting while maintaining generalization capability across diverse patient populations and disease presentations.

4 PROJECT REQUIREMENTS

4.1 Functional requirements

4.1.1 User Authentication and Management

- The system shall provide secure user registration and login functionality for patients and practitioners
- Maintain separate user profiles for patients and Ayurvedic practitioners with role-based access
- Provide password reset functionality through secure email verification

4.1.2 Constitutional Assessment Module

- The system shall present a comprehensive 35-question Prakriti assessment questionnaire
- Calculate tri-dosha scores (Vata, Pitta, Kapha) based on questionnaire responses
- The system shall classify constitutional types as single dosha dominant (>50%), bi-doshic (35-50% each), or tri-doshic (balanced 30-35%)
- The provide detailed constitutional profile reports with explanations
- Allow practitioners to override or validate computational constitutional assessments
- The system shall store constitutional assessment history for trend analysis

4.1.3 Symptom Input and Processing

- The system shall provide an intuitive interface for symptom input using checkboxes, dropdowns, and severity scales
- The system should support symptom search functionality with auto-complete suggestions
- The system shall categorize symptoms by body systems (respiratory, digestive, nervous, etc.)
- Allow severity rating for each symptom (mild, moderate, severe)
- Capture symptom duration and frequency information
- Support bulk symptom import from structured data formats

4.1.4 Disease Prediction Engine

- The system shall predict potential diseases based on symptom patterns and constitutional factors
- Provide probability scores for each predicted disease
- Rank predictions by likelihood and clinical severity
- Generate both symptom-only and constitutional-aware predictions for comparison
- Provide confidence intervals and uncertainty quantification for predictions
- The system shall support multi-label classification for co-occurring conditions

4.1.5 Expert Validation and Review

- Provide interfaces for qualified practitioners to review and validate predictions
- The system tracks expert agreement rates and inter-rater reliability metrics
- Allow experts to provide feedback and corrections on predictions
- Maintain practitioner credentials and qualification verification
- The system shall generate expert consensus reports for disputed cases
- Support collaborative case discussions among practitioners

4.1.6 Machine Learning Model Management

- The system should support multiple ML algorithms (Random Forest, SVM, Neural Networks)
- Provide model training and retraining capabilities with new data
- Implement ensemble learning with weighted voting mechanisms
- Track model performance metrics and accuracy over time
- Support testing of different model versions
- The system should provide model interpretability features showing feature importance

4.2 Non – functional Requirements

- **Scalability:** Handle large text corpora and many users.
- **Reliability:** Process noisy, code-mixed queries without frequent errors.
- **Usability:** Simple interface for both patients and practitioners.
- **Performance:** Return answers within 2–3 seconds; support 100+ concurrent users.
- **Security:** Encrypt stored data, anonymize user logs.
- **Explainability:** Show source passages, confidence scores, and reasoning paths.

5 PROJECT TIMELINE

The research project will be implemented over a 12-month period to ensure systematic progress while allowing flexibility for iterative refinement and validation. The timeline balances ambitious research goals with realistic implementation schedules based on resource availability and technical complexity requirements. The grant Chart is given below.

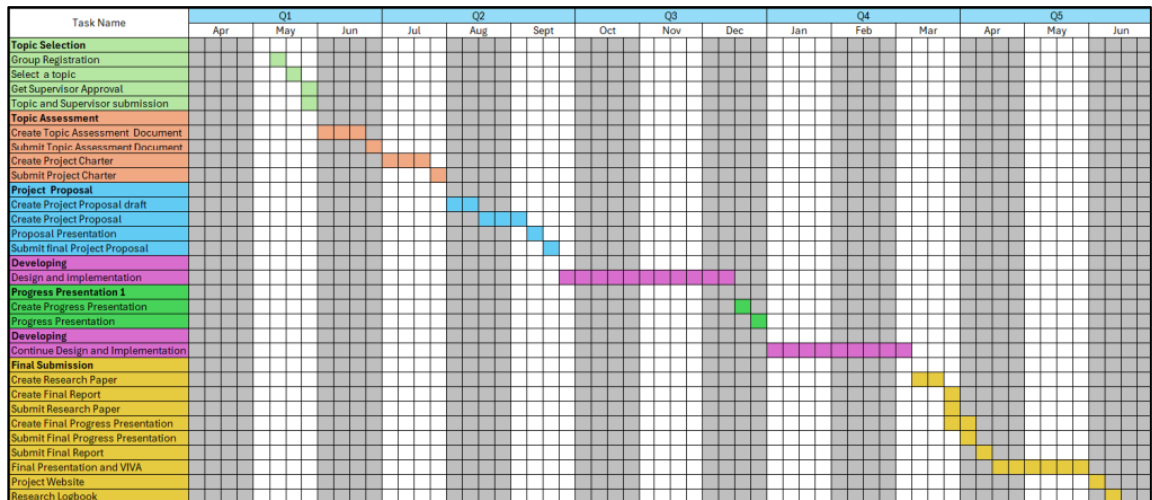


Figure 3- The Grantt Chart for the timeline

6 DESCRIPTION OF PERSONAL AND FACILITIES

Personal qualifications and technical capabilities provide strong foundation for successful completion of this ambitious interdisciplinary research project. As an undergraduate student in Information Technology with specialized focus on artificial intelligence and machine learning applications, I possess comprehensive programming expertise in Python with extensive experience in machine learning libraries including scikit-learn, TensorFlow, and pandas for data manipulation and analysis. Previous coursework in natural language processing, statistical analysis, and data mining provides essential technical foundation for Sanskrit text processing and medical terminology extraction challenges inherent in this research domain.

Background knowledge in Sanskrit linguistics and Ayurvedic principles, developed through independent study and cultural exposure, provides valuable context for understanding classical texts and traditional diagnostic approaches. This cultural and linguistic familiarity will facilitate effective collaboration with Sanskrit scholars and Ayurvedic practitioners while ensuring authentic preservation of traditional knowledge throughout the computational integration process. Strong analytical and problem-solving skills developed through academic coursework and practical projects provide capability for addressing complex technical challenges and interdisciplinary integration requirements.

Technical facilities available through SLIIT provide comprehensive computational resources essential for machine learning model development and large-scale data processing. High-performance computing laboratories equipped with modern hardware and software environments support intensive training procedures for neural network architecture and ensemble learning approaches. Access to comprehensive software development environments including Python IDEs, version control systems, and collaborative development tools enables efficient coding, testing, and documentation throughout the project lifecycle.

7 BUDGET AND BUDGET ALLOCATION

Data Collection and Processing: LKR 50,000

- Travel expenses for hospital visits and expert consultations
- Digitization and annotation costs for classical texts
- Survey distribution and data collection materials

Software and Computing Resources: LKR 25,000

- Cloud computing resources for model training
- Specialized software licenses for text processing
- Database hosting and management costs

Expert Consultation Fees: LKR 75,000

- Ayurvedic practitioner consultation fees
- Sanskrit expert annotation services
- Validation study compensation

Research Materials and Documentation: LKR 15,000

- Classical text purchases and access fees
- Reference materials and research publications
- Documentation and report preparation

Total Budget: LKR 165,000

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